

Supplementary Material for Legislator Responsiveness to Racialized Constituencies in Mexico

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A Indigenous Group Population Size Across States

	Low Population ($< 1.5\%$)	Medium Population ($1.5\% - 8.5\%$)	High Population ($> 8.5\%$)
States	Coahuila	Nayarit	Yucatan
	Tamaulipas	Baja California Sur	Chiapas
	Zacatecas	Tabasco	Guerrero
	Jalisco	Sonora	Oaxaca
	Colima	Chihuahua	Veracruz
	Sinaloa	Estado de Mexico	San Luis Potosi
	Nuevo Leon	Durango	Puebla
	Baja California	Michoacan	Hidalgo
	Queretaro	Tlaxcala	Campeche
	Guanajuato	Morelos	Quintana Roo
	Aguascalientes		
	Ciudad de Mexico		
Total States	12	10	10
Number of Legislators	238	176	213

B Political Party Coalitions in Mexico

At the time this study was conducted, the Mexican political system could be roughly divided between the ruling coalition and the opposition. The ruling coalition is composed of Movimiento Regeneración Nacional (MORENA), Partido del Trabajo (PT), Partido Verde Ecologista de México (PVEM) and Partido Encuentro Social (PES). The opposition is composed of the conservative, Partido Acción Nacional (PAN), the pre-democracy hegemonic party, the Partido Revolucionario Institucional (PRI) and the traditional left-wing party, the Partido de la Revolución Democrática, (PRD). The three of them have held a formal alliance since the 2018 election. Part of the opposition but outside their electoral coalition at the time of the experiment is Movimiento Ciudadano (MC) a center-left party.

MORENA is a political party that was founded on October 2nd, 2011, after a split within the PRD that resulted in current president Andres Manuel Lopez Obrador (AMLO) leaving that party. AMLO ran for president as a PRD candidate in 2006 and 2012, losing tight and controversial races against PAN and PRI candidates, respectively (Hanrahan and Aroch Fugellie, 2019). In the

years after, MORENA rose to power rapidly, winning the 2018 presidential election and a coalition majority of both chambers of congress. In the 2018 presidential election, AMLO obtained 53.19% of the vote as an opposition candidate. The rest of the ruling coalition consist of a Labor/workers party, the PT, “Green” Party founded in 1986 and an emerging evangelical party called PES.

MORENA’s acronym roughly translates to “dark-skinned woman.” It could be read as being called the “Brown” political party. This is both as a racial signal towards Indigenous people and darker skinned mestizos and as a religious nod, referring to the Virgin of Guadalupe, the Catholic Patron Saint of Mexico, popularly called “La Virgen Morena”. To cement his commitment to Indigenous people, AMLO took part in a Leadership Staff ceremony with Indigenous leaders from around the country after taking office. This ceremony is organized by a collective of Indigenous groups that recognize political leaders in Latin America that commit to upholding their rights (BBC News, 2018). He was also the first Mexican president to ask for forgiveness towards some indigenous groups for massacres and landgrabs by the Mexican state (El Pais, 2021).

The opposition coalition is the uneasy alliance between the main players in the pre-MORENA Mexican political system. The PAN is the most popular opposition party in Mexico. It is the only non-PRI party to have governed before MORENA, holding the presidency for two consecutive sexennial terms before ceding the executive power back to PRI’s Peña Nieto in 2012. In the present day, it is a neoliberal and socially conservative party. It is mostly identified with middle to upper class people, many of which tend to have lighter skin in Mexico (CONAPRED 2017). They were the only other party to win a majority of the vote in 2018 in 1 state to MORENA’s 31. Their candidate, Ricardo Anaya, obtained 22.27% of the vote on 2018.

Also in the opposition coalition is the PRI, the former hegemonic party in the one-party state period since the Mexican Revolution in the early 20th century up to the turn of the millennium. It was refounded with its current name in 1946. Today it is a center right party, shying away from redistributive politics during the 1960s (Hanrahan and Aroch Fugellie, 2019). Their candidate in the 2018 election, José Antonio Meade, won 16.40%The PRD, a left-wing political party, is also part of the opposition coalition. It was formed of a group of PRI leftists that were disillusioned

with the direction the revolutionary party took and founded their own opposition party (Hanrahan and Aroch Fugellie, 2019). As stated above, AMLO was a key figure in this party, particularly during the PAN governments (Hanrahan and Aroch Fugellie, 2019). Finally, MC is a center left party founded in 1999 that has risen to become the third political force in the country after the 2021 elections, behind MORENA and PAN. Neither PRD nor MC ran candidates in 2018, opting to support the PAN candidate.

C Email Content for Experiment

C.1 Spanish (Used in Audit Experiment)

“Estimado(a) Diputado(a)/Senador(a) X:

Le escribe una votante en el distrito/estado por el que usted fue electa/o. Le quiero preguntar sobre los apoyos disponibles para personas de la tercera edad en el estado. Un miembro de mi familia necesita apoyo urgentemente y no sabemos con quién acudir.

Respetuosamente, Itzayana Pech /Sofia Hernández / Julie Betancourt”

D Balance Tests for Key Covariates Across Treatment Groups

	Covariate	Chi-Squared Statistic	df	p-value
1	Gender (Male vs. Female)	0.69	2	0.71
2	Coalition (MORENA vs. Opposition)	3.15	2	0.21
3	Indigenous Population (Low, Med, High)	3.85	4	0.43

E Linear Probability Model Specifications

We estimate the treatment effects using a series of ordinary least squares (OLS) regressions with White robust standard errors. To test our hypotheses, we analyze three separate models. Our first model estimates the main treatment effects for each constituent identity group:

$$\text{Responsiveness}_i = \beta_0 + \beta_1 \text{Indigenous}_i + \beta_2 \text{Mestiza}_i + \epsilon_i \quad (1)$$

where Responsiveness_i represents our four binary outcome variables of responsiveness. Here, β_0 represents the baseline probability of response for a European-White-named constituent, the reference group, when all other predictors are at their baseline. The variable Indigenous_i is coded as 1 if the constituent's name was Indigenous-coded and 0 otherwise, with β_1 capturing the difference in response probability between Indigenous-named and European-White-named constituents. Similarly, Mestiza_i is coded as 1 if the constituent's name was Mestiza-coded and 0 otherwise, with β_2 reflecting the difference in response probability between Mestiza-named and European-White-named constituents.

Our second model includes interaction terms for Indigenous and Mestiza treatment conditions with the MORENA coalition:

$$\begin{aligned} \text{Responsiveness}_i = & \beta_0 + \beta_1 \text{Indigenous}_i + \beta_2 \text{Mestiza}_i \\ & + \beta_3 \text{MORENA Coalition}_i + \beta_4 (\text{Indigenous}_i \times \text{MORENA Coalition}_i) \\ & + \beta_5 (\text{Mestiza}_i \times \text{MORENA Coalition}_i) + \epsilon_i \end{aligned} \quad (2)$$

where $\text{MORENA Coalition}_i$ is a binary variable indicating if the legislator is part of the MORENA coalition (1 = MORENA, 0 = opposition). The coefficient β_3 represents the effect of coalition affiliation on response probability and its quality for European-White-named constituents, the baseline group. The interaction term $\text{Indigenous}_i \times \text{MORENA Coalition}_i$ measures how the DV probabilities for Indigenous-named constituents differs between MORENA and opposition legislators, with β_4 capturing this differential effect. Similarly, $\text{Mestiza}_i \times \text{MORENA Coalition}_i$ reflects the interaction between the Mestiza treatment and the MORENA coalition, where β_5 represents the difference in responsiveness to Mestiza-named constituents between MORENA and opposition legislators.

Finally, our third model examines interacts the Indigenous treatment with a categorical variable for Indigenous population size:

$$\begin{aligned} \text{Responsiveness}_i = & \beta_0 + \beta_1 \text{Indigenous}_i + \beta_2 \text{Mestiza}_i + \beta_3 \text{Indigenous Population}_i \\ & + \beta_4 (\text{Indigenous}_i \times \text{Indigenous Population}_i) \\ & + \beta_5 (\text{Mestiza}_i \times \text{Indigenous Population}_i) + \epsilon_i \end{aligned} \quad (3)$$

where $\text{Indigenous Population}_i$ is a categorical variable indicating the Indigenous population size in the legislator's district (coded as 1 = Low, 2 = Medium, 3 = High). The coefficient β_3 represents the baseline effect of Indigenous population size on responsiveness probability for European-White-named constituents. The interaction term $\text{Indigenous}_i \times \text{Indigenous Population}_i$ allows us to test whether the Indigenous treatment effect varies across different Indigenous population levels, with β_4 indicating how the responsiveness to Indigenous-named constituents changes in districts with varying Indigenous population sizes. Similarly, the interaction term $\text{Mestiza}_i \times \text{Indigenous Population}_i$ captures how the Mestiza treatment effect is moderated by Indigenous population size, with β_5 representing this interaction.

F Power Analysis for Minimum Detectable Effects

Table A: Post Hoc Power Analysis for Small Effect Size (Cohen's $h = 0.2$)

Treatment Group	Effect Size (h)	Sample Size (n)	Significance Level (α)	Power
Indigenous	0.20	192	0.05	0.500
Mestiza	0.20	207	0.05	0.530

G Leave-One-Out (Jackknife) Sensitivity Analysis

Table B: Leave-One-Out Sensitivity Analysis Across Models and Dependent Variables

DV	Model	Term	Original	Mean	SD	Min	Max	Sign Flips	Sig. Changes	Sig.
response	Main	Treatment.RecodedIndigenous	0.075	0.075	0.001	0.071	0.080	0	0	True
response	Main	Treatment.RecodedMestiza	0.016	0.016	0.001	0.011	0.020	0	0	False
response	Coalition Interaction	Treatment.RecodedIndigenous	0.063	0.063	0.001	0.052	0.072	0	6	True
response	Coalition Interaction	Treatment.RecodedMestiza	0.039	0.039	0.001	0.027	0.048	0	1	False
response	Indig Pop Interaction	Treatment.RecodedIndigenous	0.182	0.182	0.002	0.168	0.199	0	0	True
response	Indig Pop Interaction	Treatment.RecodedMestiza	0.014	0.014	0.001	-0.004	0.031	1	0	False
greeting	Main	Treatment.RecodedIndigenous	0.027	0.027	0.001	0.022	0.031	0	6	True
greeting	Main	Treatment.RecodedMestiza	0.020	0.020	0.000	0.015	0.024	0	1	False
greeting	Coalition Interaction	Treatment.RecodedIndigenous	0.024	0.024	0.001	0.012	0.024	0	0	False
greeting	Coalition Interaction	Treatment.RecodedMestiza	0.036	0.036	0.001	0.024	0.037	0	0	False
greeting	Indig Pop Interaction	Treatment.RecodedIndigenous	0.084	0.084	0.001	0.067	0.085	0	0	True
greeting	Indig Pop Interaction	Treatment.RecodedMestiza	0.018	0.018	0.001	-0.000	0.027	1	0	False
warmth	Main	Treatment.RecodedIndigenous	0.058	0.058	0.001	0.053	0.062	0	0	True
warmth	Main	Treatment.RecodedMestiza	0.015	0.015	0.000	0.010	0.019	0	1	False
warmth	Coalition Interaction	Treatment.RecodedIndigenous	0.063	0.063	0.001	0.052	0.072	0	6	True
warmth	Coalition Interaction	Treatment.RecodedMestiza	0.015	0.015	0.001	0.003	0.024	0	0	False
warmth	Indig Pop Interaction	Treatment.RecodedIndigenous	0.121	0.121	0.002	0.105	0.138	0	0	True
warmth	Indig Pop Interaction	Treatment.RecodedMestiza	-0.012	-0.012	0.001	-0.030	0.005	1	0	False
inquiry	Main	Treatment.RecodedIndigenous	0.027	0.027	0.001	0.022	0.031	0	6	True
inquiry	Main	Treatment.RecodedMestiza	0.005	0.005	0.000	0.000	0.010	0	0	False
inquiry	Coalition Interaction	Treatment.RecodedIndigenous	0.015	0.015	0.001	0.003	0.024	0	0	False
inquiry	Coalition Interaction	Treatment.RecodedMestiza	0.003	0.003	0.001	-0.009	0.012	1	0	False
inquiry	Indig Pop Interaction	Treatment.RecodedIndigenous	0.045	0.045	0.002	0.029	0.062	0	0	False
inquiry	Indig Pop Interaction	Treatment.RecodedMestiza	-0.007	-0.007	0.001	-0.026	0.010	2	0	False

To assess robustness, we conducted a leave-one-out (LOO) sensitivity analysis across all models and dependent variables (See Supplementary Appendix G for table). Each coefficient was recalculated 626 times, omitting one observation at a time, to examine how sensitive estimates were to individual cases. Overall, the main treatment effects—especially those for Indigenous-named constituents—were found to be highly stable, with no sign reversals and minimal variation. Additionally, the interaction terms involving Indigenous population size also show strong stability.

In contrast, interaction effects involving coalition affiliation exhibit moderate instability. The

Indigenous treatment and MORENA coalition interaction, for example, experiences six shifts in statistical significance across iterations, despite no directional changes. This pattern likely reflects low statistical power in subgroup analyses across both the Indigenous and Mestiza groups, reinforcing the need for caution when interpreting null results related to coalition effects. Mestiza treatment effects remain small and non-significant throughout all LOO iterations, further suggesting the absence of a detectable effect is not due to outliers but consistent with limited effect size and response rate constraints.

In sum, the LOO analysis confirms the robustness of our main findings for the indigenous treatment group and its interaction with the indigenous population interactions, while highlighting the limits of inference for the other smaller subgroups and interaction terms.

H Design and Ethics Details

This section follows the Recommended Reporting Standards for Experiments issued by the APSA Organized Section on Experimental Research. For each checklist item we state how it applies to Legislator Responsiveness to Racialized Constituencies in Mexico or note “Not Applicable (N/A)” when the item does not pertain to an e-mail audit of public officials.

H.1 Hypotheses

Four hypotheses (H1–H4) are stated and motivated in the manuscript (pp. 4-5). They were not pre-registered.

H.2 Subjects and Context

Pool and eligibility. Universe of all 626 members (499 deputies, 127 senators) of Mexico’s 65th Legislature; substitutes were contacted if the principal was on leave. Public officials therefore fell under the “public official” exemption category.

Recruitment and dates. One e-mail per legislator was sent to the official Chamber/Senate address between April 19 – April 21 2023. No follow-ups were sent.

Setting. Remote, asynchronous e-mail exchange; legislators’ offices across Mexican congressional districts.

Response-rate denominator. 626 valid addresses; all responses (4.2 %) and non-responses (95.8 %) were used in the analyses.

Survey-specific items. N/A (no survey respondents).

H.3 Allocation Method

Randomization. Legislators were block-randomized with equal probability (1/3 to each of the three treatment conditions) inside six strata defined by coalition status (governing / MORENA-led vs. opposition) crossed with state Indigenous-population tercile (1–3). Treatment labels were drawn separately within each stratum in R 4.4.1 using the `base sample()` function, producing near-balanced allocation in expectation.

H.4 Treatments

Description. A single, 88-word Spanish e-mail requesting information on benefits for the elderly signed with one of three racially coded female names (Indigenous = “Itzayana Pech”; Mestiza = “Sofía Hernández”; European-White = “Julie Betancourt”). The full text and English translation are in Appendix C.

Delivery. Manual sends from three dedicated Gmail accounts.

Exposure integrity. All messages were successfully delivered; no additional contact occurred.

Deception. Minimal deception was inherent in using fictitious constituents. The IRB judged the deception to present no more than minimal risk and waived post-experiment debriefing because subjects were public officials performing routine duties.

H.5 Results

Outcomes and covariates. Primary outcome = binary indicator of any personalized reply. Secondary outcomes = named greeting, warm tone, substantive help (all binary). Models found in Appendix E.

CONSORT flow. Because every unit is a public office (no enrollment/attrition by participants) a diagram would duplicate the simple counts already given (626 assigned → 626 delivered → 626 analyzed).

Statistical analysis. OLS with White-robust SEs; level of analysis matches level of assignment; blocking factors included. No weighting. Missingness is only outcome missingness (non-reply) and is coded 0 by design.

H.6 Other Information

IRB. Approved by the University of Southern California IRB #UP-23-00147; informed-consent requirement waived for public-official audit design.

Risk assessment. The protocol posed no more than minimal risk: e-mails were routine, content benign, and no personal data from officials were disseminated.

Confidentiality. Reply texts (which are public-record communications) are stored on encrypted drives.

Data availability. De-identified data and code are archived at Harvard Dataverse DOI 10.7910/DVN/VS3GL8.